

Burgers' N-Wave: A Correction

checking the printed formula against the pde

the claim. for $t > 0$, the problem asks whether

$$f(t, x) = \frac{x}{t} \frac{t^{-1} e^{-x^2/(4t)}}{1 + t^{-1} e^{-x^2/(4t)}} \quad \text{satisfies} \quad f_t + f f_x = f_{xx}.$$

the answer is no: the power t^{-1} in the image produces a nonzero residual. the standard n-wave uses $t^{-1/2}$ instead.

a useful identity. set $g = t^{-1} e^{-x^2/(4t)}$ and $h = 1 + g$. since $h_x = -xg/(2t)$, the given function can be written

$$f = \frac{xg}{t(1+g)} = -2 \frac{h_x}{h}.$$

writing $v = \log h$ gives $f = -2v_x$. differentiating this expression and using $h_{xx}/h = v_{xx} + v_x^2$ yields

$$\begin{aligned} f_t + f f_x - f_{xx} &= -2 \partial_x (v_t - v_{xx} - v_x^2) \\ &= -2 \partial_x \left(\frac{h_t - h_{xx}}{h} \right). \end{aligned}$$

thus the pde holds whenever h solves the heat equation $h_t = h_{xx}$.

check the printed power. direct differentiation gives

$$g_t = g \left(-\frac{1}{t} + \frac{x^2}{4t^2} \right), \quad g_{xx} = g \left(-\frac{1}{2t} + \frac{x^2}{4t^2} \right).$$

therefore $h_t - h_{xx} = -g/(2t)$, and the identity above becomes

$$\begin{aligned} f_t + f f_x - f_{xx} &= \frac{1}{t} \partial_x \left(\frac{g}{1+g} \right) \\ &= \frac{g_x}{t(1+g)^2} \\ &= \boxed{-\frac{xg}{2t^2(1+g)^2}}. \end{aligned}$$

this is nonzero whenever $x \neq 0$, so the formula in the image does not satisfy the stated pde.

the corrected n-wave. replace g by $\tilde{g} = t^{-1/2} e^{-x^2/(4t)}$ and define

$$\tilde{f}(t, x) = \frac{x}{t} \frac{t^{-1/2} e^{-x^2/(4t)}}{1 + t^{-1/2} e^{-x^2/(4t)}}.$$

now

$$\tilde{g}_t = \tilde{g} \left(-\frac{1}{2t} + \frac{x^2}{4t^2} \right) = \tilde{g}_{xx},$$

so $\tilde{h} = 1 + \tilde{g}$ satisfies $\tilde{h}_t = \tilde{h}_{xx}$. applying the identity to $\tilde{f} = -2\tilde{h}_x/\tilde{h}$ gives

$$\boxed{\tilde{f}_t + \tilde{f} \tilde{f}_x = \tilde{f}_{xx}}.$$